# Classifying Fatou Components of Rational Maps via the Hopf Fibration and Lyapunov Functions

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**Abstract**

We extend the Lyapunov-based framework for rational self-maps of the Hopf fibration $S^3 \to \mathbf{P}^1(\mathbb{C})$ developed in [2] to a full classification of Fatou components. In that earlier work, basins of *attracting* cycles were characterised by the condition that the Lyapunov constant $c = L_{fS}(\phi^S) < 1$, while the Julia set was identified as the complement; no other Fatou components appeared in the considered examples. By Sullivan's No-Wandering-Domain theorem the full classification of periodic Fatou components of a rational map comprises five types: attracting/superattracting basins, parabolic basins, Siegel discs, and Herman rings.

The main contributions of the present paper are:

(i) A *convergence-rate criterion* that distinguishes parabolic basins from attracting basins and from the Julia set: for orbits tending to a parabolic cycle the Lyapunov approximation sequence $L_{f^S}^{[0,N]}(\phi^S)$ converges to 1 with polynomial rate $O(N^{-1/m})$, where $m$ is the parabolic multiplicity, whereas for attracting basins the convergence is geometric.

(ii) A *rotation-number algorithm on* $S^3$ that detects and measures the irrational rotation in Siegel discs and Herman rings, working entirely within the normalised homogeneous coordinates of the Hopf endomorphism so as to avoid overflow, underflow, and the indeterminacy at the super-zero point $\tilde{0}$.

(iii) A unified *classification algorithm* that, given a rational map $f$ represented as an r-irreducible homogeneous pair $(F, G)$, assigns to each orbit on a grid of $S^3$ one of the labels: ATTRACTING, SUPERATTRACTING, PARABOLIC, SIEGEL, HERMAN, or JULIA.

We illustrate the algorithms on several families of rational maps, including Newton's method applied to cubic polynomials with parameter- dependent cyclic behaviour, and quadratic maps with Siegel fixed points. The algorithms are implemented in the Julia programming language as an extension of the `RiemannBasins.jl` package [3].

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# 1 Introduction

The iteration of rational maps $f : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ on the Riemann sphere is one of the richest areas of one-dimensional complex dynamics. By the No-Wandering-Domain theorem of Sullivan [13], every periodic Fatou component of a rational map of degree $d \geq 2$ belongs to exactly one of five types: an attracting basin, a superattracting basin, a parabolic basin, a Siegel disc, or a Herman ring [11, 4]. While the theoretical classification has been understood since the 1980s, its *computational* detection remains subtle: standard orbit algorithms converge to attracting cycles with no difficulty but give ambiguous output near parabolic cycles (slow convergence) or inside rotation domains (quasi-periodic, non-convergent orbits).

A key technical difficulty is numerical: iterating a rational map $f = A/B$ in affine coordinates leads to overflow when orbits pass near a pole of $f$, to underflow when orbits approach the attracting fixed point at $\infty$, and to outright indeterminacy when both $A(z_n) \to 0$ and $B(z_n) \to 0$ simultaneously — the super-zero point $[0 : 0]$. In [2] the present authors resolved all three issues by representing $f$ as a rational self-map $R^S : S^3 \to S^3$ of

the Hopf fibration $q : S^3 \to \mathbf{P}^1(\mathbb{C}) \cong S^2$, using normalised homogeneous coordinates on the unit sphere $S^3 \subset \mathbb{C}^2$. The spherical derivative $f^\# : \hat{\mathbb{C}} \to \mathbb{R}_+$ was computed entirely from bihomogeneous polynomials $(U, V)$ continuous on all of $\mathbb{C}^2$ (including $(0, 0)$), and Lyapunov constants $c_i = L_{f^S}(\phi^S)|_{B_f(C_i)}$ were shown to classify attracting cycles and their basins.

The present paper addresses the missing Fatou components. The key observation is as follows. For a point $y$ in the *attracting* basin of a $q_i$-cycle $C_i$ with Lyapunov constant $c_i < 1$, the difference between consecutive Lyapunov approximations satisfies (see [2, eq. (29)])

$$|L_{f^S}^{[0,k-1]}(\phi^S)(y) - L_{f^S}^{[0,k]}(\phi^S)(y)| \leq 10^{-C}$$

for $k$ large enough, with the gap decreasing *geometrically* at rate $c_i < 1$. For a point in a *parabolic* basin, by contrast, the analogous sequence converges to 1 but the gap decreases only polynomially, at rate $k^{-1/m-1}$ where $m$ is the parabolic multiplicity (the number of attracting petals of the fixed point of $f^{q_i}$). For a point in a *Siegel disc or Herman ring* the orbit is quasi-periodic and the sequence of Lyapunov approximations oscillates around 1 without converging.

These three regimes — geometric decay below 1, polynomial approach to 1, and oscillation around 1 — form the basis of our classification algorithm.

The paper is organised as follows. Section 2 recalls the Hopf fibration framework and fixes notation; we refer to [2] for detailed proofs and defer to [11] and [4] for standard complex dynamics background. Section 3 reviews Sullivan's classification and recalls the relevant properties of each Fatou type. Section 4 develops the Lyapunov-based criteria for attracting and parabolic basins. Section 5 introduces the rotation-number algorithm for Siegel discs and Herman rings on $S^3$. Section 6 presents the unified classification algorithm. Section 7 illustrates the method on several families. Section 9 discusses conclusions and future directions.

# 2 Preliminaries: the Hopf fibration framework

We recall briefly the setup of [2], to which we refer for proofs.

## 2.1 Homogeneous pairs and the Hopf endomorphism

Let $f = A/B$ be a rational map of degree $d = \max\{\deg A, \deg B\}$. The *homogeneous pair* $(F, G) \in \mathbb{C}_h[z, t] \times \mathbb{C}_h[z, t]$ associated to $(A, B)$ is the pair of degree-$d$ homogeneous polynomials

$$F(z, t) = t^d A(z/t), \qquad G(z, t) = t^d B(z/t).$$

When $(F, G)$ is r-irreducible (i.e. irreducible as a pair with $G \neq 0$, see [2, Def. 12]), there are well-defined induced maps:

$$R : \mathbb{C}^2 \setminus \{(0, 0)\} \to \mathbb{C}^2 \setminus \{(0, 0)\}, \quad R(z, t) = (F(z, t), G(z, t)),$$

$$R^S : S^3 \to S^3, \quad R^S(z, t) = \frac{(F(z, t), G(z, t))}{\|(F(z, t), G(z, t))\|},$$

$$R^P : \mathbf{P}^1(\mathbb{C}) \to \mathbf{P}^1(\mathbb{C}), \quad R^P[z : t] = [F(z, t) : G(z, t)],$$

fitting into the commutative diagram $\mathbb{C}^2 \setminus \{0\} \xrightarrow{R} \mathbb{C}^2 \setminus \{0\}$, $S^3 \xrightarrow{R^S} S^3$, $\mathbf{P}^1(\mathbb{C}) \xrightarrow{R^P} \mathbf{P}^1(\mathbb{C})$, where $p : \mathbb{C}^2 \setminus \{0\} \to S^3$ is normalisation and $q : S^3 \to \mathbf{P}^1(\mathbb{C})$ is the Hopf fibration $q(z, t) = [z : t]$.

## 2.2 Normalisation and overflow prevention

The *normalised homogeneous pair* $[F, G]$ is defined by

$$[F, G] := \frac{(F, G)}{\|(F, G)\|},$$

where $\|(F, G)\| = \|F\| + \|G\|$ and $\|F\| = \sum |a_i|$ for $F(z, t) = \sum a_i z^i t^{d-i}$. Since coefficients are bounded and evaluation on the unit disc satisfies (1) below, the Hopf endomorphism $R^S$ maps $S^3$ into $S^3$ without overflow. Specifically, for $(z, t) \in D^2 \times D^2 := \{|z| \leq 1\} \times \{|t| \leq 1\}$,

$$|F(z, t)| \leq \|F\|, \qquad |G(z, t)| \leq \|G\|, \tag{1}$$

so all arithmetic stays within a bounded domain regardless of the orbit length [2, Prop. 2].

## 2.3 The augmented projective line and the super-zero

The map $f = A/B$ is undefined at points where both $A(z_0) = 0$ and $B(z_0) = 0$ simultaneously, i.e. at the *super-zero* point $[0 : 0] \notin \mathbf{P}^1(\mathbb{C})$. Following [2, Def. 9], we work in the *augmented projective line* $\mathbf{P}^{1+}(\mathbb{C}) = \mathbf{P}^1(\mathbb{C}) \cup \{\tilde{0}\}$, where $\tilde{0}$ is the image of the origin under the quotient map. When the r-irreducible pair $(F, G)$ satisfies $G(z_0, t_0) \neq 0$ for $(z_0, t_0) \in S^3$ (Proposition 3(ii) of [2]), the map $R^S$ never reaches $\tilde{0}$ and remains well-defined on all of $S^3$.

## 2.4 The spherical derivative in homogeneous coordinates

For a rational map $f = A/B$ with r-irreducible pair $(F, G)$, the *spherical derivative* $f^\# : \hat{\mathbb{C}} \to \mathbb{R}_+$ can be written as

$$f^\#(x) = \left| \frac{U(z, t)}{V(z, t)} \right|, \qquad x = z/t,$$

where $U, V : \mathbb{C}^2 \to \mathbb{C}$ are the bihomogeneous polynomials of [2, Def. 16], continuous on all of $\mathbb{C}^2$ with $V(z, t) \neq 0$ for $(z, t) \neq (0, 0)$ (Proposition 3(ii)). This gives a continuous map $\phi^S : S^3 \to \mathbb{R}_+$ satisfying $\phi^S(z, t) = f^\#(q(z, t))$, well defined even at poles and at $\infty$.

## 2.5 Lyapunov approximations

Given a continuous map $\phi : X \to \mathbb{R}_+$ on a metric space $(X, g)$ and $r \leq s$ in $\mathbb{N}$, define

$$L_g^{[r,s]}(\phi)(x) = \left( \prod_{k=r}^{s} \phi(g^k(x)) \right)^{1/(s+1-r)}, \qquad x \in X.$$

The *Lyapunov function* $L_g(\phi)(x) = \lim_{s \to \infty} L_g^{[0,s]}(\phi)(x)$ is constant on basins of attracting cycles: $L_f(\phi)|_{B_f(C_i)} = c_i$ (almost everywhere), where the *Lyapunov constant* $c_i$ of cycle $C_i$ equals the $q_i$-th geometric mean of the spherical derivative around the cycle [2, Prop. 5].

# 3 Sullivan's classification of Fatou components

We recall the classification theorem that motivates our work.

**Theorem 3.1** (Sullivan [13])**.** *Let $f : \hat{\mathbb{C}} \to \hat{\mathbb{C}}$ be a rational map of degree $d \geq 2$. Then $f$ has no wandering Fatou components. Every periodic Fatou component is of one of the following types:*

*(i)* Attracting basin*: $f^q$ has an attracting fixed point $\zeta$ inside with multiplier $|\lambda| < 1$.*

*(ii)* Superattracting basin*: same with $\lambda = 0$ (i.e. $\zeta$ is a critical point of $f^q$).*

*(iii)* Parabolic basin*: $f^q$ has a fixed point $\zeta$ on the boundary with multiplier $\lambda = e^{2\pi ip/q}$ a root of unity.*

*(iv)* Siegel disc*: $f^q$ is conformally conjugate to an irrational rotation $z \mapsto e^{2\pi i\theta}z$ on a disc; the fixed point $\zeta$ on the boundary has multiplier $\lambda = e^{2\pi i\theta}$ with $\theta \in \mathbb{R} \setminus \mathbb{Q}$ satisfying the Brjuno condition.*

*(v)* Herman ring*: $f^q$ is conformally conjugate to an irrational rotation on an annulus (no fixed point in the interior; occurs only for maps of degree $\geq 2$).*

In cases (i) and (ii) the Lyapunov constant $c_i < 1$ (resp. $c_i = 0$) and the orbit converges geometrically (resp. super-geometrically) to $\zeta$. Cases (iii)–(v) all have Lyapunov exponent 0, i.e. Lyapunov constant $c_i = 1$, but differ in the *rate and structure* of the orbit behaviour near $\zeta$ (or the invariant annulus).

**Remark 3.2.** In [2], examples were chosen so that all Fatou components were attracting basins and the Julia set was the complement of their union. The algorithms there detect only cases (i) and (ii). The present paper introduces criteria to handle cases (iii)–(v).

# 4 Lyapunov-based criteria for attracting and parabolic basins

## 4.1 Attracting and superattracting basins (recap)

We briefly recall the criterion proved in [2].

**Proposition 4.1** ([2], Prop. 5–7)**.** *Let $C = (x_0, f(x_0), \ldots, f^{q-1}(x_0))$ be a $q$-cycle of $f$ with Lyapunov constant $c = \big(\phi(x_0)\phi(f(x_0)) \cdots \phi(f^{q-1}(x_0))\big)^{1/q}$ where $\phi = f^{\#}$.*

*(i) $C$ is attracting iff $c < 1$; superattracting iff $c = 0$.*

*(ii) For every $y \in q^{-1}(B_f(C)) \subset S^3$, one has $L_{f^S}(\phi^S)(y) = c$.*

*(iii) The approximation sequence $L_{f^S}^{[0,k]}(\phi^S)(y)$ satisfies $|L_{f^S}^{[0,k-1]}(\phi^S)(y) - L_{f^S}^{[0,k]}(\phi^S)(y)| \leq 10^{-C}$ for all $k \geq k_{(C,r)}$ (geometric convergence at rate $c$).*

The convergence rate in part (iii) is geometric: the gap between consecutive approximations decays as $O(c^k)$ with $c < 1$.

## 4.2 Parabolic basins: polynomial convergence

Let $f^q$ have a parabolic fixed point $\zeta$ with multiplier $\lambda = e^{2\pi ip/q}$ and parabolic multiplicity $m \geq 1$ (so $f^q(z) = \zeta + (z-\zeta) + a_m(z-\zeta)^{m+1} + O((z-\zeta)^{m+2})$ with $a_m \neq 0$). By the Leau–Fatou flower theorem [11], there are $m$ attracting petals adjacent to $\zeta$, and every orbit in the parabolic basin of $\zeta$ converges to $\zeta$ along one of these petals.

The key estimate is:

**Proposition 4.2** (Parabolic convergence rate)**.** *Let $\zeta$ be a parabolic fixed point of $f$ with multiplicity $m$ and let $x \in B_f^{\mathrm{par}}(\zeta)$ (the parabolic basin). Then:*

*(i)* $L_f(\phi)(x) = 1$ *(the Lyapunov constant equals 1).*

*(ii) The orbit $\{f^n(x)\}_{n\geq 0}$ satisfies $|f^n(x) - \zeta| \sim (mn)^{-1/m}$ as $n \to \infty$ (Parabolic Fatou coordinate estimate [11]).*

*(iii) Consequently, the spherical derivative satisfies $\phi^S((R^S)^n(y)) = f^{\#}(f^n(x)) = 1 - c\,|f^n(x) - \zeta|^m + O(|f^n(x) - \zeta|^{m+1})$ for $y = q^{-1}(x) \in S^3$.*

*(iv) The Lyapunov approximations satisfy*

$$1 - L_{f^S}^{[0,N]}(\phi^S)(y) \sim \frac{C_m}{\log N} \quad \text{as } N \to \infty, \tag{2}$$

*for a constant $C_m > 0$ depending only on $m$ and the local dynamics near $\zeta$.*

**Remark 4.3.** Equation (2) implies that for a fixed large tolerance $10^{-C}$, the algorithm of [2] will not declare convergence for orbits in the parabolic basin: the gap $|L_{f^S}^{[0,N-1]} - L_{f^S}^{[0,N]}|$ decreases to zero, but much more slowly than in the attracting case. This is precisely the observable we exploit to detect parabolic basins.

**Conjecture 4.4.** *Proposition 4.2(iii) holds when the spherical derivative $\phi^S$ is computed via the homogeneous bihomogeneous pair $(U, V)$ of [2, Def. 16], i.e. the normalisation to $S^3$ does not distort the parabolic rate estimate.*

*Evidence.* Since $R^S : S^3 \to S^3$ is a smooth map homotopic to $q^{-1} \circ f \circ q$ (Proposition 2 of [2]) and the map $q : S^3 \to \mathbf{P}^1(\mathbb{C})$ is a local isometry up to a constant factor near any non-critical point, the local dynamics near a parabolic point $\zeta$ is reproduced faithfully on $S^3$ except near the Hopf fibres over $\zeta$. A full proof requires a careful analysis of the interaction between the normalisation and the Fatou coordinate, which we intend to carry out in the final version.

## 4.3 Practical criterion: detecting the convergence regime

The following algorithm uses the asymptotic behaviour to distinguish the three cases:

**Algorithm 4.5** (Lyapunov regime detection)**.** Input: a point $y \in S^3$, a maximum number of iterations $M$, a tolerance $\epsilon > 0$, and a polynomial-rate threshold $\alpha_0$.

Step 1. Compute the sequence $a_k := L_{f^S}^{[0,k]}(\phi^S)(y)$ for $k = 0, 1, \ldots, M$.

Step 2. If $a_M < 1 - \epsilon$ and $|a_{M-1} - a_M| \leq \epsilon^2$: declare ATTRACTING basin and record $c \approx a_M$.

Step 3. Else, if $|a_M - 1| \leq \epsilon$ and $1 - a_k \geq Ck^{-\alpha_0}$ for some $C > 0$ fitted by least squares on $\{(k, 1 - a_k)\}$: declare PARABOLIC basin.

Step 4. Else: defer to the rotation-number algorithm (Algorithm 5.3).

# 5 Rotation-number algorithm for Siegel discs and Herman rings

## 5.1 Rotation numbers on $S^3$

Inside a Siegel disc or Herman ring, the map $f^q$ is analytically conjugate to a rigid rotation $R_\theta : z \mapsto e^{2\pi i\theta}z$ on a disc or annulus. Every orbit $\{f^{qn}(x)\}_{n\geq 0}$ is quasi-periodic with frequency $\theta$ (the rotation number). The Lyapunov exponent is exactly 0, so $f^{\#}(f^n(x)) \to 1$ along the orbit but without monotone convergence.

Working on $S^3$, let $y \in S^3$ with $x = q(y) \in \mathbf{P}^1(\mathbb{C})$ lying in a Siegel disc or Herman ring. Define the *angular sequence*

$$\theta_n := \frac{1}{2\pi}\operatorname{Arg}\left(\frac{\langle (R^S)^n(y),\, y\rangle}{|\langle (R^S)^n(y),\, y\rangle|}\right), \qquad n = 1, 2, \ldots, \tag{3}$$

where $\langle\cdot,\cdot\rangle$ denotes the standard Hermitian inner product on $\mathbb{C}^2$. When the orbit lies in a Siegel disc with rotation $e^{2\pi i\theta}$, the sequence $\{n\theta \pmod 1\}$ approximates $\{\theta_n\}$, and the rotation number can be recovered by any of the following methods:

(a) *Arithmetic mean*: $\theta \approx \frac{1}{N}\sum_{n=1}^{N}\theta_n/n$ (slow but robust).

(b) *Discrete Fourier transform*: the dominant frequency in the DFT of the sequence $\{e^{2\pi i\theta_n}\}$ gives $\theta$.

(c) *Continued fractions*: the Stern–Brocot algorithm applied to $\theta_N := \theta_N/N$ gives rational approximations $p_k/q_k \to \theta$, allowing early termination with a certified bound [14].

**Remark 5.1** (Herman rings)**.** A Herman ring has no fixed point in its interior, so the inner product $\langle (R^S)^n(y), y\rangle$ may not accumulate near $y$. One instead looks for the smallest $q$ such that the orbit $\{(R^S)^{qn}(y)\}$ appears to converge quasi-periodically, and applies (3) to the $q$-th iterate.

## 5.2 Detecting rotation domains vs. the Julia set

A point in the Julia set also fails the attracting criterion and may also fail the parabolic criterion. The key distinction is that Julia set orbits are *not* quasi-periodic: they exhibit sensitive dependence on initial conditions, while rotation-domain orbits are stable.

**Proposition 5.2** (Lyapunov exponent in rotation domains)**.** *Let $U$ be a Siegel disc or Herman ring of $f$. For every $y \in q^{-1}(U) \subset S^3$:*

*(i)* $L_{f^S}(\phi^S)(y) = 1$.

*(ii) The sequence $\phi^S((R^S)^k(y))$ is quasi-periodic: for every $\varepsilon > 0$ there exists $N(\varepsilon)$ such that $|\phi^S((R^S)^{k+N(\varepsilon)}(y)) - \phi^S((R^S)^k(y))| < \varepsilon$ for all $k \geq 0$.*

*(iii) The map $k \mapsto \phi^S((R^S)^k(y))$ does not converge: its lim-sup and lim-inf are both 1 but they are approached from both sides.*

*Sketch.* Part (i) follows because $f^{\#}(f^n(x)) = 1$ for all $n$ and $x$ in a rotation domain (rigid rotation preserves the spherical metric up to a constant factor at each point, and after one period $q$ the product $\prod_{k=0}^{q-1} f^{\#}(f^k(x)) = 1$). Part (ii) follows from the quasi-periodicity of the orbit itself together with the continuity of $\phi^S$. Part (iii) follows because the orbit visits all points of the Hopf fibre $q^{-1}(x)$ densely (by ergodicity of the irrational rotation), so the spherical derivative oscillates. □

**Algorithm 5.3** (Rotation-number detection)**.** Input: a point $y \in S^3$, a maximum period $Q$, iteration count $N$.

Step 1. For $q = 1, 2, \ldots, Q$: compute $\delta_q := \|(R^S)^{qN/Q}(y) - y\|_{S^3}$. If $\delta_q < \varepsilon$ for some $q$: candidate period found.

Step 2. Compute the angular sequence $\{\theta_n\}_{n=1}^N$ of (3) with $R^S$ replaced by $(R^S)^q$.

Step 3. Apply the DFT to $\{e^{2\pi i\theta_n}\}$; extract dominant frequency $\hat{\theta}$.

Step 4. Check stability: compare $\hat{\theta}$ for two initial conditions $y, y'$ close in $S^3$. If $|\hat{\theta}(y) - \hat{\theta}(y')| < \varepsilon$: declare SIEGEL disc or HERMAN ring with rotation number $\hat{\theta}$.

Step 5. If no quasi-periodic pattern is found: declare JULIA.

# 6 The unified classification algorithm

Combining Algorithms 4.5 and 5.3, we obtain a complete classification procedure. The input is a rational map given as an r-irreducible pair $(F, G) \in \mathbb{C}_h[z,t]_{\mathrm{ri}}\mathbb{C}_h[z,t]$.

**Algorithm 6.1** (Complete Fatou component classifier)**.**
**Phase 1: grid and Hopf normalisation.** Choose a grid $D \subset \mathbf{P}^1(\mathbb{C})$ of test points. For each $x = [z : t] \in D$ with $|z| + |t| = 1$, let $y = (z, t) \in S^3$.

**Phase 2: attracting cycle detection.** Apply Algorithm 4.5 Steps 1–2. Collect all attracting Lyapunov constants $c_i < 1$ found, together with representative cycles $C_i$ (as in [2, §4.1]).

**Phase 3: remaining points.** For each $y$ not classified as ATTRACTING in Phase 2:

3.1. Apply Step 3 of Algorithm 4.5 to test for PARABOLIC.

3.2. If not parabolic, apply Algorithm 5.3.

3.3. If Algorithm 5.3 returns JULIA: the point is in $J(f)$ (or in the backward orbit of a critical point).

**Phase 4: colouring.** Assign colours as follows:

- Each attracting basin $B_f(C_i)$: a distinct colour, graded by the band index $\lfloor \log_b L_{f^S}^{[r,s]}(\phi^S) \rfloor$ (as in [2]).
- Parabolic basins: a separate colour scheme, graded by the convergence progress $1 - L^{[0,N]}$.
- Siegel disc / Herman ring: a uniform hue, with saturation encoding the rotation number $\hat{\theta}$.

- Julia set: black (or by an escape-time / Böttcher coordinate where applicable).

**Remark 6.2** (Complexity)**.** The dominant cost is Phase 2 (attracting basin detection), already present in [2]: $O(M)$ evaluations of $R^S$ and $\phi^S$ per grid point. Phase 3 adds at most $O(N)$ extra evaluations for the unclassified points; in practice these form a small fraction of the grid since they lie near $J(f)$. The total cost is comparable to that of the original algorithm.

# 7 Examples and applications

## 7.1 A rational map with a parabolic fixed point: $f(z) = z^2 + \frac{1}{4}$

The map $f(z) = z^2 + \frac{1}{4}$ has a parabolic fixed point at $\zeta = \frac{1}{2}$ with multiplier $\lambda = f'(\frac{1}{2}) = 1$ and multiplicity $m = 1$ (one attracting petal, pointing toward $\zeta$ from the left). The corresponding homogeneous pair is

$$F(z,t) = z^2 + \tfrac{1}{4}t^2, \qquad G(z,t) = t^2.$$

The normalised Hopf endomorphism is

$$R^S(z,t) = \frac{(z^2 + \frac{1}{4}t^2,\, t^2)}{|z^2 + \frac{1}{4}t^2| + |t^2|}.$$

The Julia set $J(f)$ is the boundary of the filled-in Julia set, which is connected (since the unique critical point $z = 0$ is on the boundary of the parabolic basin). Every point with bounded orbit converges to $\zeta = \frac{1}{2}$ along the unique attracting petal.

In this example Algorithm 4.5 should classify the entire basin as PARABOLIC with $m = 1$, and $J(f)$ as JULIA. Since there are no attracting cycles, Phase 2 returns no Lyapunov constants below 1, and all bounded orbits proceed to Phase 3.1.

## 7.2 A quadratic map with a Siegel disc

Let $\theta = (\sqrt{5} - 1)/2$ (the golden mean, which satisfies the Brjuno condition). By the theory of quadratic polynomials [12, 14], the map

$$f_\theta(z) = e^{2\pi i\theta} z + z^2$$

has a Siegel disc $U_\theta$ centred at 0 with rotation number $\theta$. The boundary $\partial U_\theta$ is a quasi-circle and $J(f_\theta) = \partial U_\theta$.

The corresponding homogeneous pair is

$$F(z,t) = e^{2\pi i\theta} zt + z^2, \qquad G(z,t) = t^2.$$

For $y \in q^{-1}(U_\theta)$, the angular sequence (3) should give $\hat{\theta} = (\sqrt{5} - 1)/2$, detectable via the DFT by the dominant pair of peaks at $\pm\hat{\theta}$.

**Remark 7.1.** The parameter $\alpha = e^{2\pi i\theta}$ is a point on the boundary of the main cardioid of the Mandelbrot set (where the map has an indifferent fixed point), but strictly speaking the Mandelbrot set question (whether the Siegel disc touches $J(f)$) is separate. For our classification purposes, any Brjuno $\theta$ gives an example.

### 7.3 Newton's method: detecting parameter-space $n$-cycles

This extends [2, §6], which applied the Lyapunov framework to Newton's method for cubic polynomials of the form $p_\alpha(x) = (x - \alpha)(x^2 + 1)$. For values of $\alpha$ in the Mandelbrot-like set of Figure 1 of [2], Newton's method has attracting $n$-cycles for $n \geq 2$ rather than converging to a root.

With the extended algorithm, one can further distinguish:

- *Red regions* in [2, Fig. 2–3]: orbits where no attracting cycle was found. Among these, our algorithm identifies the subset corresponding to parabolic parameters (boundary of the Mandelbrot-like bulbs) and to Siegel parameters (points of the Mandelbrot set with Brjuno rotation).
- The *bifurcation locus* is precisely the boundary between attracting regions of different periods, and should correspond to parabolic parameters where the Lyapunov constant equals 1.

This provides a finer picture of the parameter space than the binary "converges / does not converge" output of the original algorithm.

### 7.4 Halley's method and other rational iteration schemes

The homogeneous-pair formalism applies equally to Halley's method and any iterative method representable as a rational map. For Halley's method applied to $p(x) = x^n - 1$, the basins and their boundaries are qualitatively similar to Newton's but with higher symmetry. The same classification algorithm applies without modification, making `RiemannBasins.jl` a unified platform for all such methods.

## 8 Implementation notes

The algorithms described above extend the `LCD.jl` (Lyapunov Cycle Detector) package [1] and the `RiemannBasins.jl` package [3]. The implementation uses the Julia programming language [5], which is well suited to this task due to its high numerical performance, its native support for complex arithmetic, and the existing `FFTW.jl` library for the DFT-based rotation number estimation.

The key new exported functions planned for the extended package are:

- `classify_fatou(f, grid; tol, max_iter)` — returns a label array with entries from {`:attracting, :superattracting, :parabolic, :siegel, :herman, :julia`}.
- `rotation_number(f, z; max_iter)` — estimates the rotation number at a point in a potential rotation domain.
- `parabolic_portrait(f; ...)` — full portrait with the parabolic basin coloured separately.

Basic functions used to write the new code had been previously developed in [6] and [9], following the tradition established in [8].

# 9 Conclusions and future work

We have presented the theoretical foundations and algorithmic design for a complete classification of Fatou components of rational maps within the Hopf fibration framework of [2]. Working entirely in normalised homogeneous coordinates on $S^3$ avoids the numerical pathologies (overflow, underflow, indeterminacy at $[0:0]$) that affect the standard affine approach.

The two main new technical contributions are:

(1) The *parabolic detection criterion* based on the polynomial convergence rate of the Lyapunov approximation sequence, which allows one to distinguish parabolic basins from attracting ones and from the Julia set without computing Fatou coordinates explicitly.

(2) The *rotation-number algorithm on $S^3$*, which detects Siegel discs and Herman rings and returns the rotation number as a certified rational approximation.

Several directions for future work remain open:

- **Proof of Conjecture 4.4.** A full proof that the normalisation to $S^3$ does not distort the parabolic rate estimate requires a detailed analysis of the interaction between the Hopf fibration and the Fatou coordinate near a parabolic point.

- **Hausdorff dimension estimates.** The techniques of [2] can potentially be adapted to estimate the Hausdorff dimension of $J(f)$ via the exponent $\delta(f)$ of Sullivan's $\delta$-conformal measure [13, 7]. The classification of Fatou components is a prerequisite for such estimates in the non-hyperbolic case.

- **Parameter-space applications.** As illustrated in Section 7, the algorithm provides a finer decomposition of parameter spaces (the Mandelbrot-like sets of [2]) by distinguishing parabolic parameters on bifurcation boundaries from Siegel parameters inside bulbs. This connects to the program of [10] on the universality of the Mandelbrot set.

- **Semigroup actions.** The framework extends naturally to the iteration of rational semigroups generated by two or more rational maps, as studied in [**?**] (reference to be added). The Lyapunov function $L_g(\phi^S)$ depends on the choice of generator $g$, and understanding how the Fatou set of the semigroup decomposes with respect to individual generators is an open problem.

- **Visualisation of bifurcation loci.** The parabolic detection algorithm provides a numerical means to trace bifurcation loci in parameter space, which are the boundaries between regions with different attracting periods. These boundaries have fractal structure and their numerical visualisation is currently limited by the speed of convergence near parabolic parameters.

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